

Brushless Motor

BLV Series R Type

OPERATING MANUAL

Installation and Connection Edition

Before using the product

Safety precautions

Precautions for use

Checking the product

Installation

Connection

Guidance

Inspection and maintenance

Appendix

Thank you for purchasing an Oriental Motor product.

This Operating Manual describes product handling procedures and safety precautions.

- Please read it thoroughly to ensure safe operation.
- Always keep the manual where it is readily available.

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1 Before using the product

Only qualified personnel of electrical and mechanical engineering should work with the product.
Use the product correctly after thoroughly reading the section “Safety precautions”. In addition, be sure to observe the contents described in warning, caution, and note in this manual.
The product described in this manual is designed and manufactured to be incorporated in general industrial equipment. Do not use for any other purpose.
For the power supply, use a DC power supply with reinforced insulation on its primary and secondary sides. Oriental Motor Co., Ltd. is not responsible for any compensation for damage caused through failure to observe this warning.

■ Overview of the product

- **Compact and light weight**
Compact and lightweight are achieved, offering the dimensions of 65 mm × 75 mm × 29 mm (2.56 in. × 2.95 in. × 1.14 in.) and the mass of 0.12 kg (0.26 lb.).
- **Compatible with battery drive**
This product is compatible with the allowable operating voltage of 15 VDC to 55 VDC, taking into account voltage fluctuations.
- **Stability at low speeds**
Excellent stability at low speeds is obtained in comparison with conventional brushless motors. Using this product can achieve smooth starting and stopping of equipment.
- **Equipped with the power removal function**
The power removal function is a function that stops supplying the power to the motor by the hardware. The power removal function is assumed to be used to prevent unexpected starting of the moving parts of equipment when an operator works inside the operating range of the moving parts.
- **Compatible with network communications**
Parameters can be set and operation can be executed or stopped via communications of RS-485 (Modbus RTU) or CAN (CANopen).
Parameters can also be set using the **MEXE02**.

■ Related operating manuals

Operating manuals are not included with the product. Download them from Oriental Motor Website Download Page or contact your nearest Oriental Motor sales office. To use the product, read this manual together with the **BLV Series R Type OPERATING MANUAL Function Edition**.

Operating manual name
BLV Series R Type OPERATING MANUAL Installation and Connection Edition (this document)
BLV Series R Type OPERATING MANUAL Function Edition
BLV Series R Type Driver CANopen Communication Profile
BLV Series R Type Motor OPERATING MANUAL

Search for an operating manual by the model name shown on the nameplate.

2 Safety precautions

The precautions described below are intended to ensure the safe and correct use of the product, and to prevent the customer and others from exposure to the risk of injury. Use the product only after carefully reading and fully understanding these instructions.

Description of signs

 WARNING	Handling the product without observing the instructions that accompany a "WARNING" symbol may result in serious injury or death.
 CAUTION	Handling the product without observing the instructions that accompany a "CAUTION" symbol may result in injury or property damage.
	The items under this heading contain important handling instructions that the user should observe to ensure safe use of the product.
	The items under this heading contain related information and contents to gain a further understanding of the text in this manual.

Explanation of graphic symbols



Indicates "prohibited" actions that must not be performed.



Indicates "compulsory" actions that must be performed.

 WARNING	
	• Do not use the product in explosive or corrosive environments, in the presence of flammable gases, in places subjected to splashing water, or near combustibles. Doing so may result in fire, electric shock, or injury. • Do not transport, install, connect or inspect the product while the power is supplied. Always turn off the power before carrying out these operations. This may result in electric shock or damage to equipment. • Do not touch the driver while the power is on. Doing so may result in fire or electric shock. • Do not forcibly bend, pull, or pinch the cable. Doing so may result in fire, electrical shock, or damage to equipment. • Do not touch the motor or driver when conducting the insulation resistance measurement or dielectric strength test. Accidental contact may result in electric shock. • Do not disassemble or modify the driver. Doing so may result in electric shock, injury, or damage to equipment. Refer all such internal inspections and repairs to the branch or sales office from which you purchased the product.
	• Only qualified and educated personnel should be allowed to perform installation, connection, operation and inspection/troubleshooting of the product. Handling by unqualified and uneducated personnel may result in fire, electric shock, injury, or damage to equipment. • Use a motor and a driver only in the specified combination. An incorrect combination may cause fire, electric shock, or damage to equipment.

WARNING

	• If the alarm function (protective function) of the driver is activated, remove the cause before resetting the alarm. Continuing the operation without removing the cause of the problem may result in malfunction of the motor, leading to injury or damage to equipment. • Install the driver in an enclosure. Inappropriate installation may result in injury. • Always keep the power supply voltage of the driver within the specified range. Failure to do so may result in fire, electric shock, or damage to equipment. • Connect the cables securely according to the wiring example. Failure to do so may result in fire, electric shock, or damage to equipment. • For a main power supply, I/O signals power supply and a power supply for communication, use a battery or a DC power supply with reinforced insulation on its primary and secondary sides. Failure to do so may result in electric shock. • Turn off the main power supply in the event of a power failure. Otherwise, the motor may suddenly start when the power is restored, causing injury or damage to equipment. • Always turn off the power before performing maintenance or inspection. Failure to do so may result in electric shock.
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CAUTION

	• Do not use the driver beyond its specifications. Doing so may result in fire, electric shock, injury, or damage to equipment. • Do not touch the driver while operating or immediately after stopping. The surface is hot, and this may cause a skin burn(s). • Keep the area around the driver free of combustible materials. Failure to do so may result in fire or a skin burn(s). • Do not leave anything around the driver that would obstruct ventilation. Doing so may result in damage to equipment.
	• Securely install the driver to the mounting plate. Inappropriate installation may cause the motor or driver to detach and fall, resulting in injury or damage to equipment. • Make sure the wiring for the power supply does not disconnect. This may result in damage to equipment. • When moving the moving part by hands, put the motor into a non-excitation state. Operating in a state where the motor is excited may cause injury. • Pay enough attention to safe operation when starting and stopping the motor by switching ON-OFF of the power supply. Failure to do so may result in injury or damage to equipment. • Be sure to ground the driver to prevent them from being damaged by static electricity. Failure to do so may result in fire or damage to equipment. • Provide an emergency stop device or emergency stop circuit external to the equipment so that the entire equipment will operate safely in the event of a system failure or malfunction. Failure to do so may result in injury. • Immediately when a problem occurred, stop operation and turn off the main power supply. Failure to do so may result in fire, electrical shock, or injury.

3 Precautions for use

This section covers restrictions and requirements the user should consider when using the product.

■ Wiring

● Connecting a motor and a driver

Use the dedicated connection cable (sold separately) when extending the wiring distance between the motor and the driver.

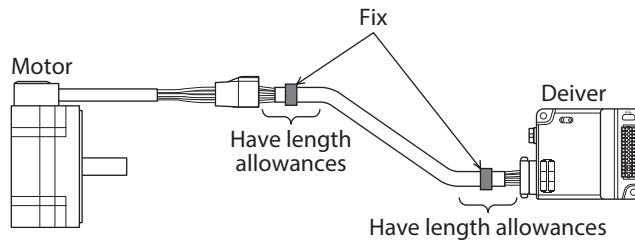
● Notes when the cable is used

Note the following points when the cable is used.

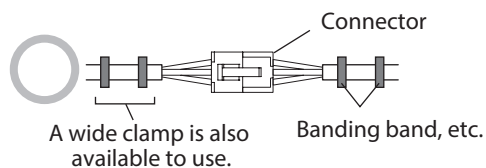
How to fix the cable

Fix the cable at the positions near the connector so that no stress due to the bending or self-weight of the cable is applied on the connector.

Also, do not excessively bend the cable near the connection part of the connector. Applying stress on the cable may cause poor contact or disconnection, leading to malfunction or heat generation.

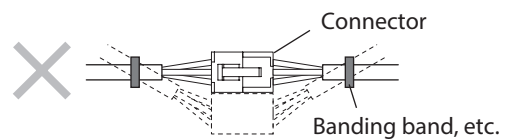


● Fixing at two places on each side



Fix using two cable ties or a wide clamp.

● Fixing at one place on each side

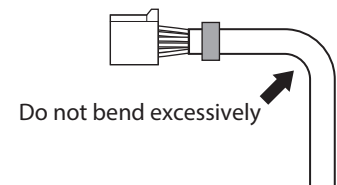


When the cable is moved, it causes the connectors to move, causing stress to apply on the connector part.

When bending the cable

Do not excessively bend the cable.

Applying stress on the cable may cause poor contact or disconnection, leading to malfunction or heat generation.



● Note when connecting a power supply whose positive terminal is grounded

The USB connector on the driver is not electrically insulated. When grounding the positive terminal of the power supply, do not connect any equipment (PC, etc.) whose negative terminal is grounded. Doing so may cause the driver and these equipment to short, damaging both.

● Noise elimination measures

Refer to p.25 for noise elimination measures.

■ Static electricity measures

The driver uses components that are sensitive to static electricity. Take measures against static electricity since it may cause the driver to malfunction or suffer damage.

■ Insulation resistance measurement and dielectric strength test

Do not conduct the insulation resistance measurement or the dielectric strength test with the motor and driver connected.

Conducting the insulation resistance measurement or dielectric strength test with the motor and driver connected may result in damage to the product.

■ Operations

● Regeneration energy

When the motor is used in operation such as rapid acceleration/deceleration of an inertia body or vertical drive (elevating equipment), regeneration energy may generate. Since the driver has no function to consume regeneration energy, if the output capacity or overvoltage allowance of the DC power supply is small, the protective function for the power supply or driver may be activated to stop the motor. When performing these operations, use a DC power supply or battery that has a large output capacity or overvoltage allowance.

If the protective function for the power supply or driver is activated, contact your nearest Oriental Motor sales office.

■ Power removal function

- When the power removal function is used, be sure to conduct a risk assessment of equipment in advance and check that the safety requirements of the safety-related parts of a control system are satisfied.
- The design of the safety-related parts of a control system using the power removal function should be performed by qualified personnel who are trained in the relevant safety standards and understand the contents of this chapter.
- If the power removal function is activated, the motor output shaft may be rotated by external forces (gravity on a vertical axis, etc.). To hold the motor output shaft in position, install an external brake mechanism or equivalent. The brake mechanism of the electromagnetic brake motor is used for the purpose to hold the position. Do not use the brake mechanism of the electromagnetic brake motor for braking the motor rotation. This may result in injury or damage to equipment.
- If the power removal function is activated, the driver stops supplying the power to the motor. However, the power supplying to the driver is not shut off, and the driver is not electrically insulated either. Before performing maintenance or inspection, always turn off the driver power, and check the PWR/SYS LED is turned off. Failure to do so may result in electric shock.
- If the inverter circuit is failed, the motor output shaft may rotate up to 180 degrees in an electrical angle (36 degrees in a mechanical angle) even when the power removal function is activated. Make sure this movement does not cause hazardous situations. Failure to do so may result in injury or damage to equipment.
- Connect the I/O signals related to the power removal function to an external device which conforms to the safety standard.
- Be sure to perform the verification testing of the power removal function when starting up or maintaining the equipment, or when replacing the driver. Failure to do so may result in injury or damage to equipment. If the power removal function is used in an incorrect state such as incorrect wiring of I/O signals, the power removal function may not be activated properly, causing hazardous situations.

■ Saving the data

● Notes when saving the data to the non-volatile memory

Do not turn off the power supply while writing the data to the non-volatile memory, and also do not turn off for five seconds after the completion of writing the data. Doing so may abort writing the data and cause an alarm of EEPROM error to generate.

The non-volatile memory can be rewritten approximately 100,000 times.

4 Checking the product

4-1 Package contents

Verify that the items listed below are included.
Report any missing or damaged items to the branch or sales office from which you purchased the product.

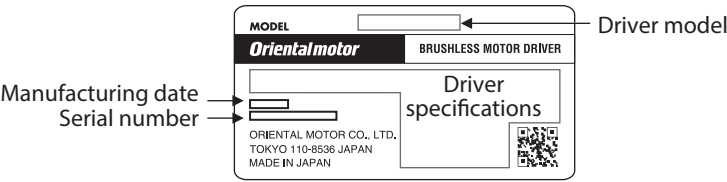
- ☐ Driver 1 unit
- ☐ Instructions and Precautions for Safe Use 1 copy

4-2 How to identify the product model

BLVD - **K** **R** **D**
 1 2 3

1	Driver type	BLVD: BLV Series driver
2	Power supply input	K: DC power input
3	Type	R: R type

4-3 Information about nameplate



4-4 Products possible to combine

Verify the model name of the purchased product against the model shown on the nameplate.
For details on the combination of a motor and a gearhead, refer to the operating manual of the motor.

- The box (□) in the model name indicates a number representing the gear ratio.
- The box (◆) in the model name indicates **F** or **B** representing the cable outlet direction.

■ Motor-parallel shaft gearhead

Output power	Motor model	Gearhead model	Driver Model
60 W	BLMR460SHK-GFV	GFV4G□	BLVD-KRD
100 W	BLMR5100K-GFV-◆	GFV5G□	
200 W	BLMR5200K-GR-◆	5GR□	
	BLMR6200SK-GFV-◆	GFV6G□	
400 W	BLMR6400SK-GFV-◆		

■ Motor-hollow shaft flat gearhead

Output power	Motor model	Gearhead model	Driver Model
60 W	BLMR460SHK-GFV	GFS4G□FR	BLVD-KRD
100 W	BLMR5100K-GFV-◆	GFS5G□FR	
200 W	BLMR5200K-GR-◆	5GR□FR	
	BLMR6200SK-GFV-◆	GFS6G□FR	
400 W	BLMR6400SK-GFV-◆		

■ CS geared motor

Output power	CS geared motor Model	Driver Model
60 W	BLMR260HK-□CS	BLVD-KRD

■ Round shaft type

Output power	Motor model	Driver Model
60 W	BLMR260HK-A	BLVD-KRD
100 W	BLMR5100K-A-◆	
200 W	BLMR5200K-A-◆	
	BLMR5200K-K-◆	
400 W	BLMR5400K-A-◆ BLMR5400K-K-◆	

■ Motor-parallel shaft gearhead with electromagnetic brake

Output power	Motor model	Gearhead model	Driver Model
100 W	BLMR5100KM-GFV-◆	GFV5G□	BLVD-KRD
200 W	BLMR5200KM-GR-◆	5GR□	
	BLMR6200SKM-GFV-◆	GFV6G□	
400 W	BLMR6400SKM-GFV-◆		

■ Motor-hollow shaft flat gearhead with electromagnetic brake

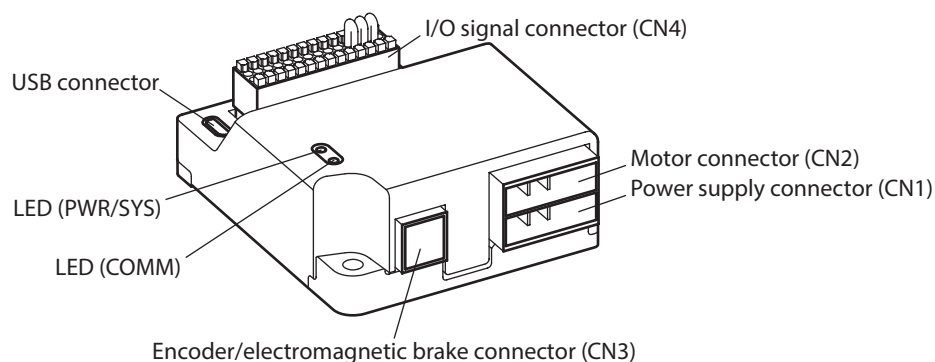
Output power	Motor model	Gearhead model	Driver Model
100 W	BLMR5100KM-GFV-◆	GFS5G□FR	BLVD-KRD
200 W	BLMR5200KM-GR-◆	5GR□FR	
	BLMR6200SKM-GFV-◆	GFS6G□FR	
400 W	BLMR6400SKM-GFV-◆		

■ Round shaft type with electromagnetic brake

Output power	Motor model	Driver Model
100 W	BLMR5100KM-A-◆	BLVD-KRD
200 W	BLMR5200KM-A-◆	
	BLMR5200KM-K-◆	
400 W	BLMR5400KM-A-◆	
	BLMR5400KM-K-◆	

4-5 Names and functions of parts

This section explains the name and function for each part of the driver.



Name	Sign	Description
Power supply connector	CN1	Connects the power supply cable.
Motor connector	CN2	Connects the motor connector.
Encoder/electromagnetic brake connector	CN3 ENC/MB	Connects the encoder/electromagnetic brake connector.
I/O signal connector	CN4 I/O	Connects with external equipment.
USB connector		Connects a PC in which the MEXE02 has been installed.
LED	PWR/SYS	Refer to "4-6 Indication of LEDs."
	COMM	

4-6 Indication of LEDs

The driver status and the communication status via RS-485 or CAN can be checked using the indication of LEDs.

■ PWR/SYS LED

The status of the driver can be checked.

LED status	Description
No light	The main power is not supplied.
White light	The main power is supplied. (PWR)
Blinking red	An alarm is being generated. The alarm type generated can be checked by counting the number of times the LED blinks. The LED will be lit in white when the alarm is reset.
Blinking white	The power removal function has been activated. The LED will be lit in white when the power removal function and the ETO status is released.
Blinking blue	Information is being generated. The LED will be lit in white when the information is cleared.
Repeating "Green → Red → Simultaneously lit (yellow) → No light"	This is the driver simulation mode.

■ COMM LED

The communication status can be checked. It is unlit in the initial state.

When each communication is started, the COMM LED is lit or blinks depending on the communication status.



If communications of CAN and RS-485 were used simultaneously, CAN communication is prioritized.

● When using via RS-485 communication

LED status	Description
White light or blinking white	The driver communicates with the client station properly via RS-485 communication. (C-DAT)
Red light	An error occurs in communication with the client station via RS-485 communication. The LED will be lit or blink in white when the communication status returns to the normal state. (C-ERR)

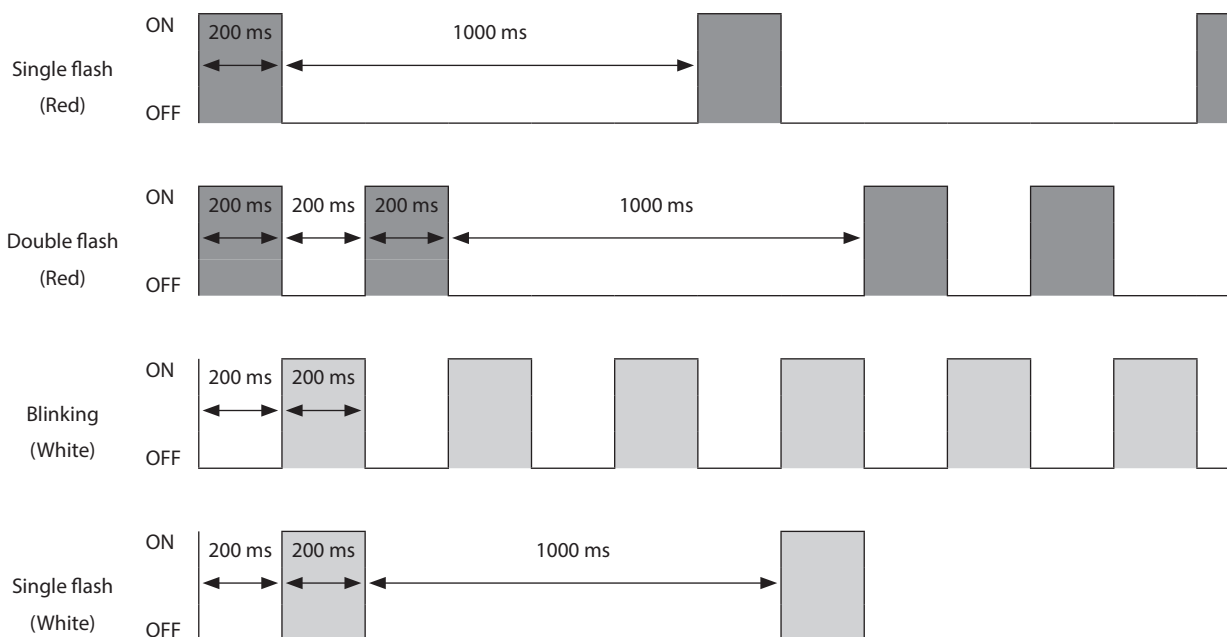
● When using via CAN communication

With the COMM LED, the red LED (Error LED) and the white LED (Run LED) are indicated independently for each function.

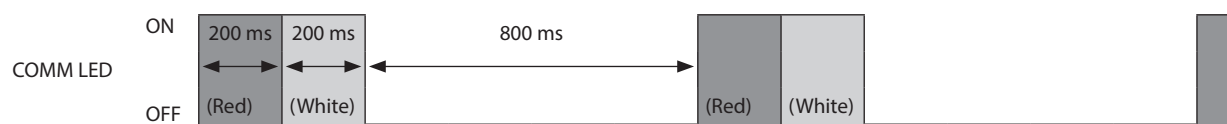
Red LED (Error LED): Indicates the status of CAN communication and the error in CAN messages.

White LED (Run LED): Indicates the status of CANopen network state machine.

LED status		Description
Red LED (Error LED)	No light	The driver is in a state of normal operation.
	Single flash	The error counter of the CAN controller reached the warning level.
	Double flash	Guarding event or Heartbeat event has occurred.
	Lighting	The driver is in a state of Bus off.
White LED (Run LED)	Blinking	The driver is in a state of PRE-OPERATIONAL.
	Single flash	The driver is in a state of STOPPED.
	Lighting	The driver is in a state of OPERATIONAL.

Blinking state of LED

An example when the red LED (Error LED) and the white LED (Run LED) blink simultaneously is shown below.

Example: When the red LED is "Single flash" and the white LED is "Single flash"

The red LED is prioritized in a state where the red LED and the white LED blink simultaneously.

5 Installation

5-1 Installation location

The driver is designed and manufactured to be incorporated in equipment. Install it in a well-ventilated location that provides easy access for inspection. The location must also satisfy the following conditions:

- Inside an enclosure that is installed indoors (provide vent holes)
- Operating ambient temperature: 0 to +40 °C [+32 to +104 °F] (non-freezing)
- Operating ambient humidity: 85 % or less (non-condensing)
- Area free of explosive atmosphere, toxic gas (such as sulfuric gas), or liquid
- Area not exposed to direct sun
- Area free of excessive amount of dust, iron particles or the like
- Area not subject to splashing water (rain, water droplets), oil (oil droplets) or other liquids
- Area free of excessive salt
- Area not subject to continuous vibration or excessive shocks
- Area free of excessive electromagnetic noise (from welders, power machinery, etc.)
- Area free of radioactive materials, magnetic fields or vacuum
- Up to 1000 m (3300 ft.) above sea level

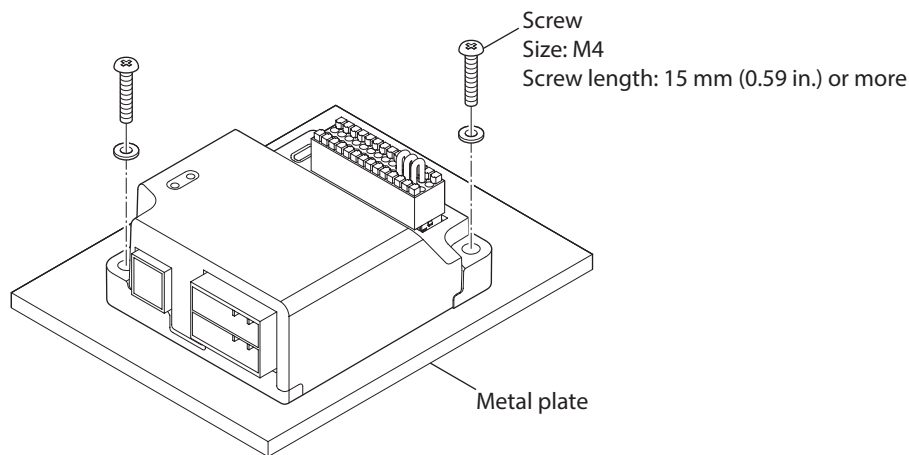
5-2 How to install

The driver can be installed in any direction.

Install the driver to a flat metal plate* offering high heat conductivity.

When installing the driver, use the mounting holes and secure to the metal plate with two screws (M4: not included).

* Material: Aluminum, size: 200×200×2 mm (7.87×7.87×0.08 in.) or equivalent



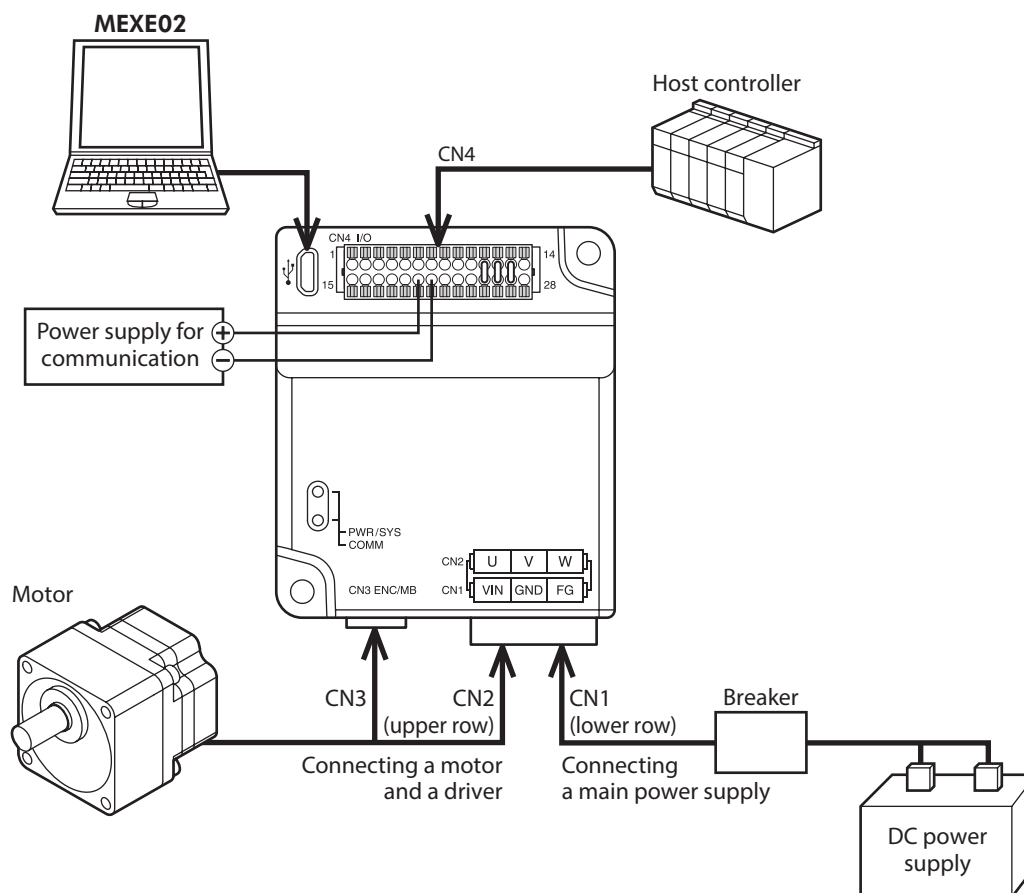
Note

- Install the driver in an enclosure.
- Do not install any equipment that generates a large amount of heat or noise near the driver.
- Do not install the driver underneath a host controller or other equipment vulnerable to heat.
- If the ambient temperature of the driver exceeds the upper limit of the operating ambient temperature, reconsider the ventilation condition or forcibly cool the area around the driver using a fan in order to keep within the operating ambient temperature.

6 Connection

This chapter explains how to connect the driver with the motor, power supplies, and I/O signals.

6-1 System configuration



Note

- When connecting, pay attention to the polarity of the power supply. Reverse-polarity connection may cause damage to the driver.
- Connect the connectors securely. Insecure connector connections may cause malfunction or damage to the product.
- Keep 3.5 m (11.5 ft.)* or less for the wiring distance between a motor and a driver. Extending the wiring distance to 3.5 m (11.5 ft.)* or more may result in heat generation from the driver or increase of the electrical noise emitted from the products including the motor and the cable.

* 3 m (9.8 ft.) for the connector type

memo

- Before connecting or disconnecting a connector, turn off the main power supply and the power supply for communication, and check the PWR/SYS LED has been turned off.
- When wiring the I/O signal cable, provide a clearance at least 100 mm (3.94 in.) from inductive loads such as electromagnetic relay, and do not parallel to the power supply cable and connection cable.
- Do not wire the power supply cable in the same cable duct with other power line or motor cable.

6-2 Connecting the main power supply (CN1)

The power supply current capacity varies depending on the motor connected.
Insert the connector of the power supply cable into the main power supply connector (CN1) on the driver.
The power supply cable **LC03D06A** (sold separately) is provided.

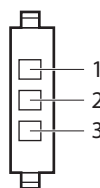
Motor	Input power supply voltage	Power supply current capacity
60 W	24 VDC	5.5 A or more
	48 VDC	3 A or more
100 W	24 VDC	10 A or more
	48 VDC	5 A or more
200 W	24 VDC	18 A or more
	48 VDC	9 A or more
400 W	48 VDC	16 A or more

Note

- When connecting, pay attention to the polarity of the power supply. Connection with incorrect polarity may cause damage to the driver.
- The input current varies depending on the main power supply voltage or the motor output power. Select a wire diameter suitable to the input current.
- To connect the power supply, use as thick a cable as possible and wire over the shortest possible distance. If a thin cable is used or the wiring distance is long, the voltage drop is increased.
- Do not wire the power supply cable in the same cable duct with other power line or motor cable.

■ Pin assignment

Pin No.	Name	Description
1	VIN	Main power supply input (24 to 48 VDC)
2	GND	Ground for main power supply
3	FG	Frame ground

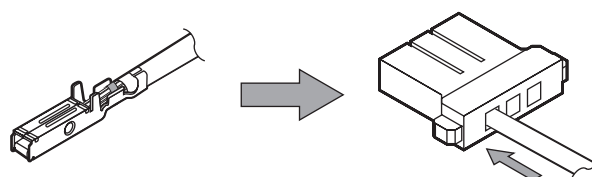


Connector: F32FSS-03V-KX (J.S.T. Mfg. Co., Ltd.)
Contact: LF3F-41GF-P2.0 (J.S.T. Mfg. Co., Ltd.)
Designated crimp tool: YRF-880 (J.S.T. Mfg. Co., Ltd.)

■ How to wire the connector

Applicable lead wire: AWG18 to AWG16 (0.75 to 1.25 mm²)

1. Strip the insulation of the lead wires.
2. Crimp the lead wires and contacts using the designated crimp tool.
3. Insert the lead wires that have crimped contacts into the connector for CN1.



6-3 Connecting the motor and the driver (CN2, CN3)

Insert the connectors of the motor cable or connection cable into the motor connector (CN2) and the encoder/electromagnetic brake connector (CN3) on the driver.

Use the dedicated connection cable (sold separately) when extending the wiring distance between the motor and the driver.

The maximum extension distance including the cable length of the motor itself should be 3.5 m [11.5 ft.] (3 m [9.8 ft.] for the connector type).

Note

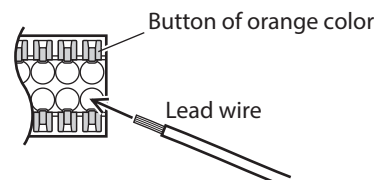
- Connect the connectors securely. Insecure connector connections may cause malfunction or damage to the product.
- Be sure to insert and pull out the connector while holding the connector part. Do not apply any force in a direction other than the direction of inserting and pulling out the connector. Applying improper force may cause damage to the product.
- Do not lift up the product by holding the motor cable or connection cable. Doing so may result in damage to the product.

6-4 Connecting the I/O signals (CN4)

Connect the I/O signal cable, power supply cable for communication, RS-485 communication cable, or CAN communication cable to CN4 according to your method for using.

■ How to wire the CN4 connector

- Applicable lead wire: AWG26 to 20 (0.14 to 0.5 mm²)
 - Lead wire strip length: 7 mm (0.28 in.)
1. Strip the insulation of the lead wire.
 2. Insert the lead wire while pushing the button of the orange color with a screwdriver.
 3. After having inserted, release the button to secure the lead wire.



Connector: DFMCO,5/14-ST-2,54
(PHOENIX CONTACT GmbH & Co. KG)



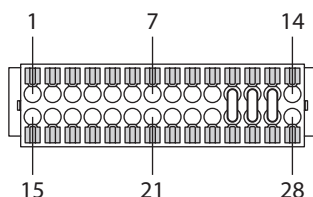
- Be certain the I/O signal cable is as short as possible.
- Use the CAN-Bus cable for the CAN communication cable.

■ Pin assignments list



- All input signals of the driver are photocoupler inputs.
- The status of signals is shown as follows.
I/O signals for normally open: "ON: Current-carrying" "OFF: Not current-carrying"
I/O signals for normally closed: "ON: Not current-carrying" "OFF: Current-carrying"

Pin No.	Signal name	Description*
1	OUT0+	Control output 0 (COMM-PWR)
2	OUT0-	
3	OUT1+	Control output 1 (ALM-B)
4	OUT1-	
5	CAN_L	CAN Low
6	CAN_H	CAN High
7	CAN_GND	Ground for CAN communication
8	485GND	Ground for RS-485 communication
9	TR+	Positive side of signal for RS-485 communication
10	TR-	Negative side of signal for RS-485 communication
11	HWT01+	Positive side of power removal input 1
12	HWT01-	Negative side of power removal input 1
13	0V	0 V for internal connection
14	EDM-	Negative side of power removal failure monitoring output



* Values in parentheses () are initial values.

Pin No.	Signal name	Description*
15	IN-COM	Common for IN0 to IN3 inputs
16	IN0	Control input 0 (ID-SEL0)
17	IN1	Control input 1 (ID-SEL1)
18	IN2	Control input 2 (STOP)
19	IN3	Control input 3 (FREE)
20	NET-VIN	Power supply for communication
21	NET-GND	Ground for power supply for communication
22	485GND	Ground for RS-485 communication
23	TR+	Positive side of signal for RS-485 communication
24	TR-	Negative side of signal for RS-485 communication
25	+V	Positive side for internal connection
26	HWT02+	Positive side of power removal input 2
27	HWT02-	Negative side of power removal input 2
28	EDM+	Positive side of power removal failure monitoring output

* Values in parentheses () are initial values.

Note

- "No.7: CAN_GND," "No.8: 485GND," "No.21: NET-GND," and "No.22: 485GND" are connected to a signal ground (SG). The SG is insulated from "No.13: 0 V" and "Ground for main power supply."
- If the power removal function is not used, connect a jumper wire (included) between the terminals as shown in the figure.
- When connecting, pay attention to the polarity of the power supply. Connection with incorrect polarity may cause damage to the driver.

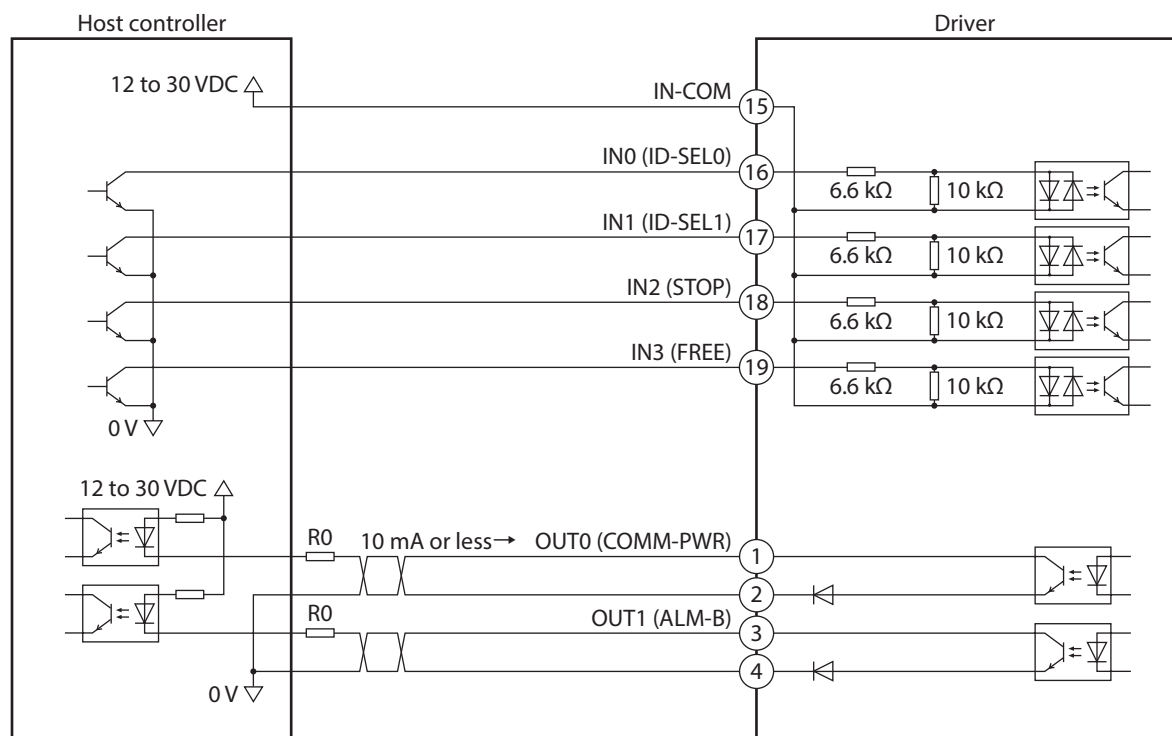
■ Power supply current capacity of power supply for communication

Power supply current capacity	Input power supply voltage
0.2 A or more	24 to 48 VDC

Note

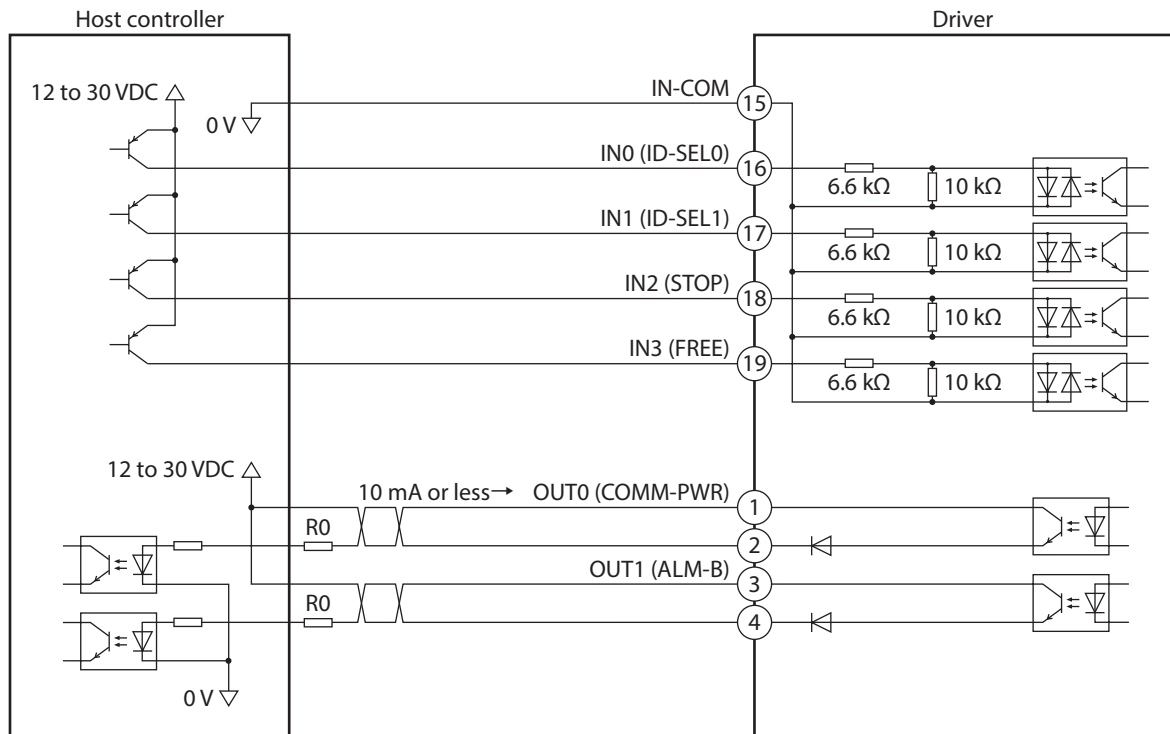
Be sure to connect a power supply for communication when using RS-485 communication or CAN communication.

■ Connection example with a current sink output circuit



* Values in parentheses () are initial values.

■ Connection example with a current source output circuit

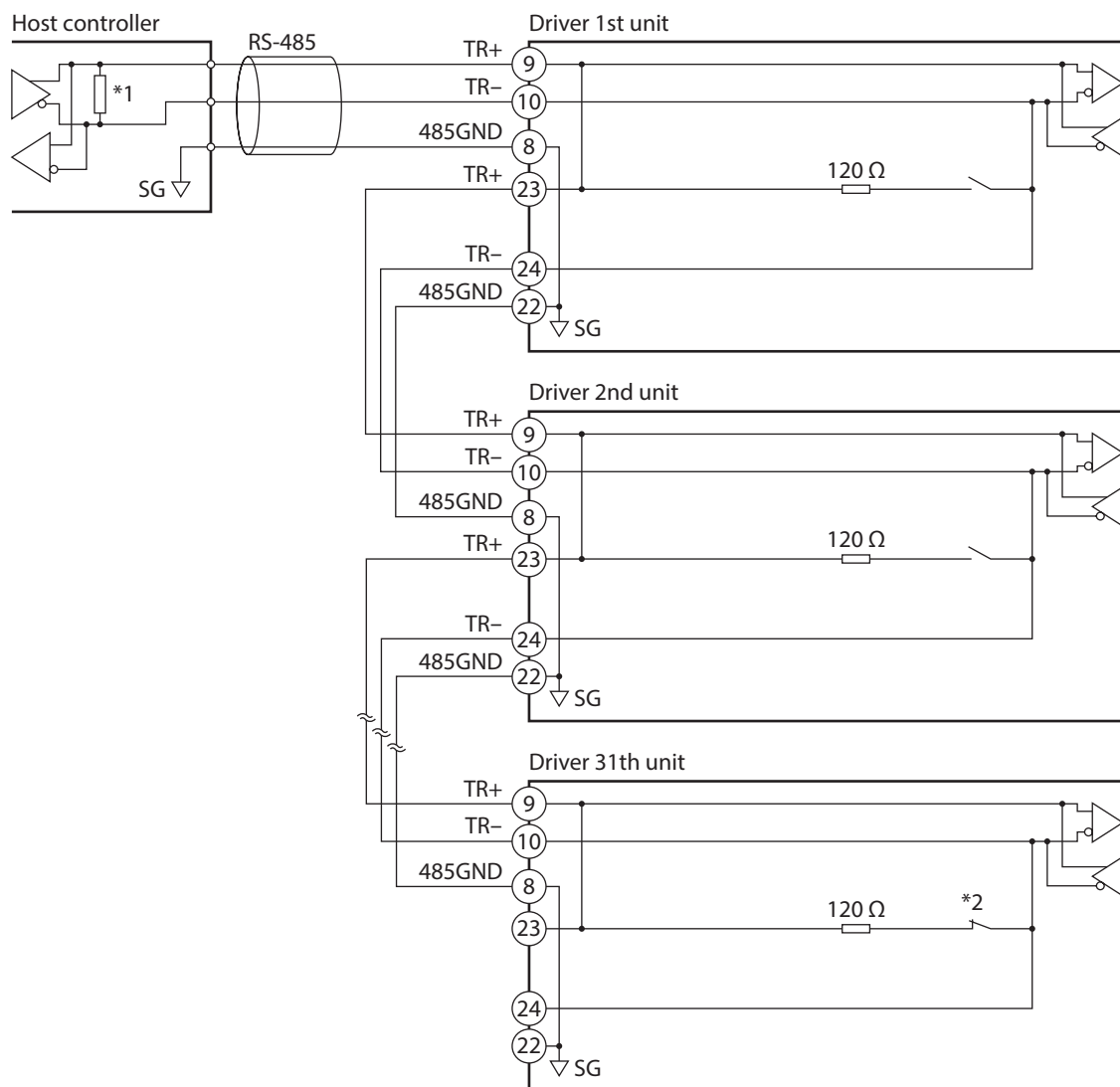


* Values in parentheses () are initial values.

Note

- Use input signals at 12 to 30 VDC.
- Use output signals at 12 to 30 VDC, 10 mA or less. If the current exceeds 10 mA, connect an external resistor R0 so that the current becomes 10 mA or less.
- The saturated voltage of output signals is 2.0 VDC maximum.

■ Connection example with a host controller (RS-485 communication)



*1 Termination resistor 120 Ω

*2 Set the "RS-485 communication termination resistor" parameter to "Enable" with the **MEXE02**.

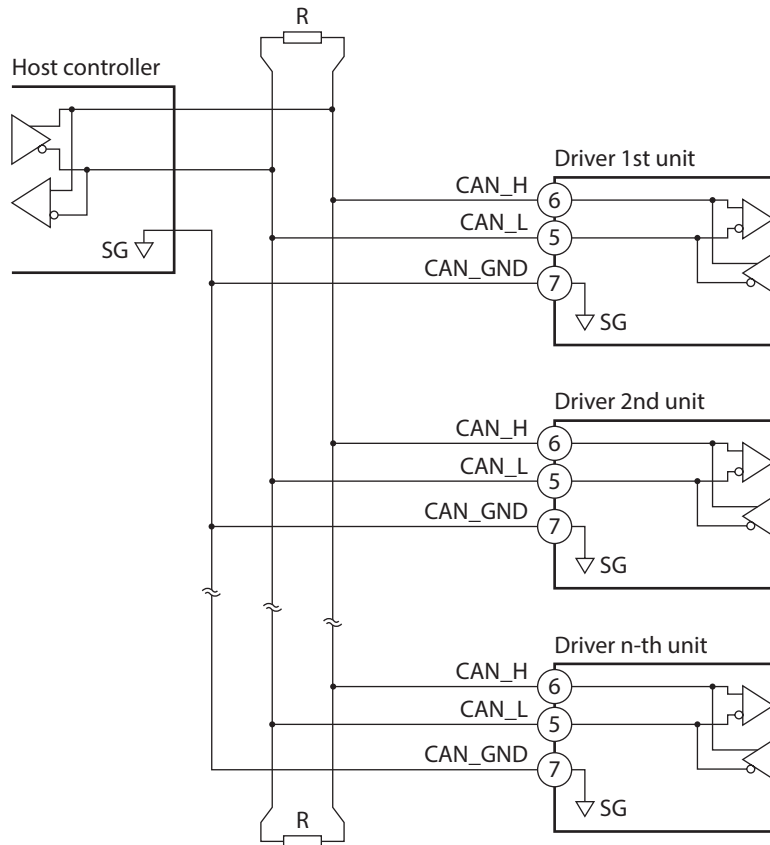
Note

- Be sure to connect a power supply for communication when using the product via RS-485 communication.
- Use twisted-pair wires for the communication cable and keep the total extension distance up to 10 m (32.8 ft.).
- Keep 31 units or less for the number of drivers connected.
- The SG is insulated from the ground for main power supply.

memo

Refer to p.36 for the communication specifications.

■ Connection example with a host controller (CAN communication)



R: Termination resistor

Connect the termination resistors (120 Ω , 1/4 W or more) on both ends of a bus. Termination resistors are not included with the product.

Note

- Be sure to connect a power supply for communication when using the product via CAN communication.
- Use the CAN-Bus cable for the CAN communication cable.
- The SG is insulated from the ground for main power supply.

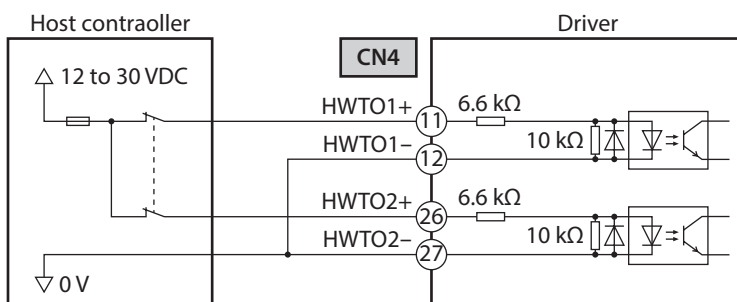
memo

Refer to p.37 for the communication specifications.

■ Connection example with a host controller (power removal function)

● Input signal

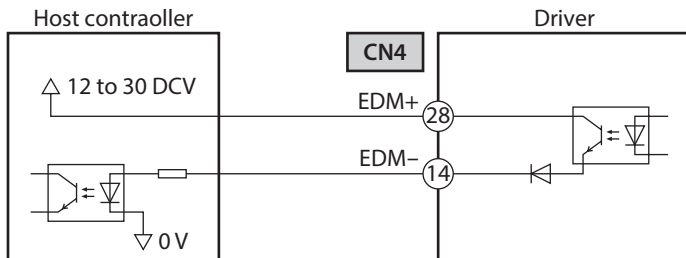
Signal name	Specifications
HWT01+ input HWT01- input	Input voltage: 12 to 30 VDC
HWT02+ input HWT02- input	



Provide individual contacts for operating the HWT01 input and the HWT02 input.

● Output signal

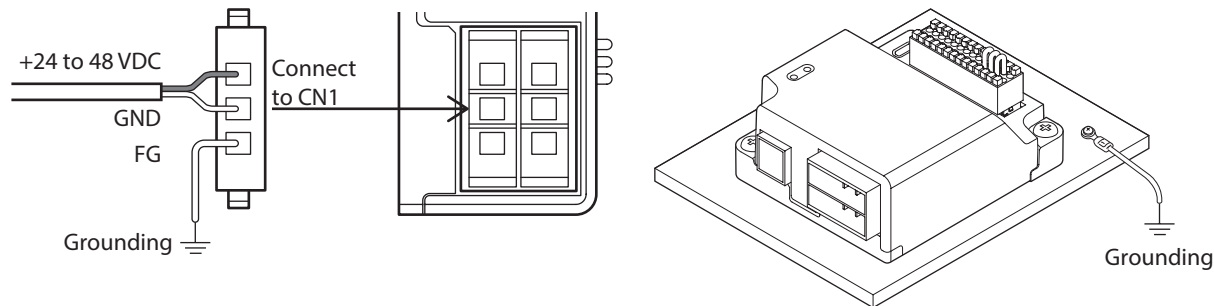
Signal name	Specifications
EDM+ output EDM- output	Voltage: 12 to 30 VDC Current: 10 mA or less Output saturated voltage: 2.0 V maximum



The EDM output is not an output signal to ensure the safety. Do not use the EDM output for any other purpose except for monitoring a failure.

6-5 **Grounding**

Ground the FG terminal of the main power supply connector. Install the driver to a metal surface that has grounded. Do not share the grounding wire with a welder or any other power equipment.



Note Static electricity may cause damage to the products if they are not grounded.

6-6 **Connecting the USB cable**

Connect the USB cable to the USB connector when using the **MEXE02**.

Specifications of USB cable

Specification	USB2.0 (full speed)
Cable	Length: 3 m (9.8 ft.) or less Shape: A to micro B

- Note**
- Connect the driver and PC directly with the USB cable without using a hub or an extension cable.
 - In large electrically noisy environments, use the USB cable with a ferrite core or install a ferrite core to the USB cable.
 - The driver's USB connector is not electrically insulated. When grounding the positive terminal of the power supply, do not connect any equipment (PC, etc.) whose negative terminal is grounded. Doing so may cause the driver and these equipment to short, damaging both.

6-7 Noise elimination measures

There are two types of electrical noise: One is noise that enters the driver from the outside and causes it to malfunction, and the other is noise that is emitted from the driver and causes peripheral equipment to malfunction. For noise entering the driver from the outside, take measures to prevent the driver from malfunctioning. It is necessary to take appropriate measures because the signal lines are very likely to be affected by the noise. For the noise that is emitted by the driver, take measures to suppress it.

■ Measures to eliminate electrical noise

- Install the driver inside a metal enclosure.
- Use a DC power supply that complies with the EMC Directive/Regulations.
- When relays or electromagnetic switches are used, use noise filters or CR circuits to suppress the surge generated by these devices.
- Keep the cables as short as possible.
- Use shielded twisted pair cables for power lines and signal lines.
- Keep power lines, such as motor and power supply cables, at least 100 mm (3.94 in.) away from signal lines. Also, do not bundle the cables or wire them in parallel. If the power cables and signal cables must cross, cross them at right angles.
- Wrap the cable around a ferrite core. This prevents the propagated noise from entering the driver or from being emitted from the driver. To increase the effect of noise attenuation by the ferrite core, wrap the cable several more turns.
- To ground a shielded cable, use a conductive cable clamp that ensures contact with the entire circumference of the shielded cable, and ground it as close to the equipment as possible.
- Use the shortest, thickest grounding wire possible.
- Choose a large and uniformly conductive surface for the grounding point.
- Ground the motor, driver and other peripheral control equipment so that a potential difference does not occur among the grounds.

■ Oriental Motor noise suppression products

● Surge suppressors

This product is effective in suppressing the surge that occurs in a contact part of the relay or electromagnetic switch. Connect it when using a relay or electromagnetic switch. Check on the Oriental Motor Website for the model of the CR circuit for surge suppression.

6-8 Compliance with the EMC Directive/Regulations

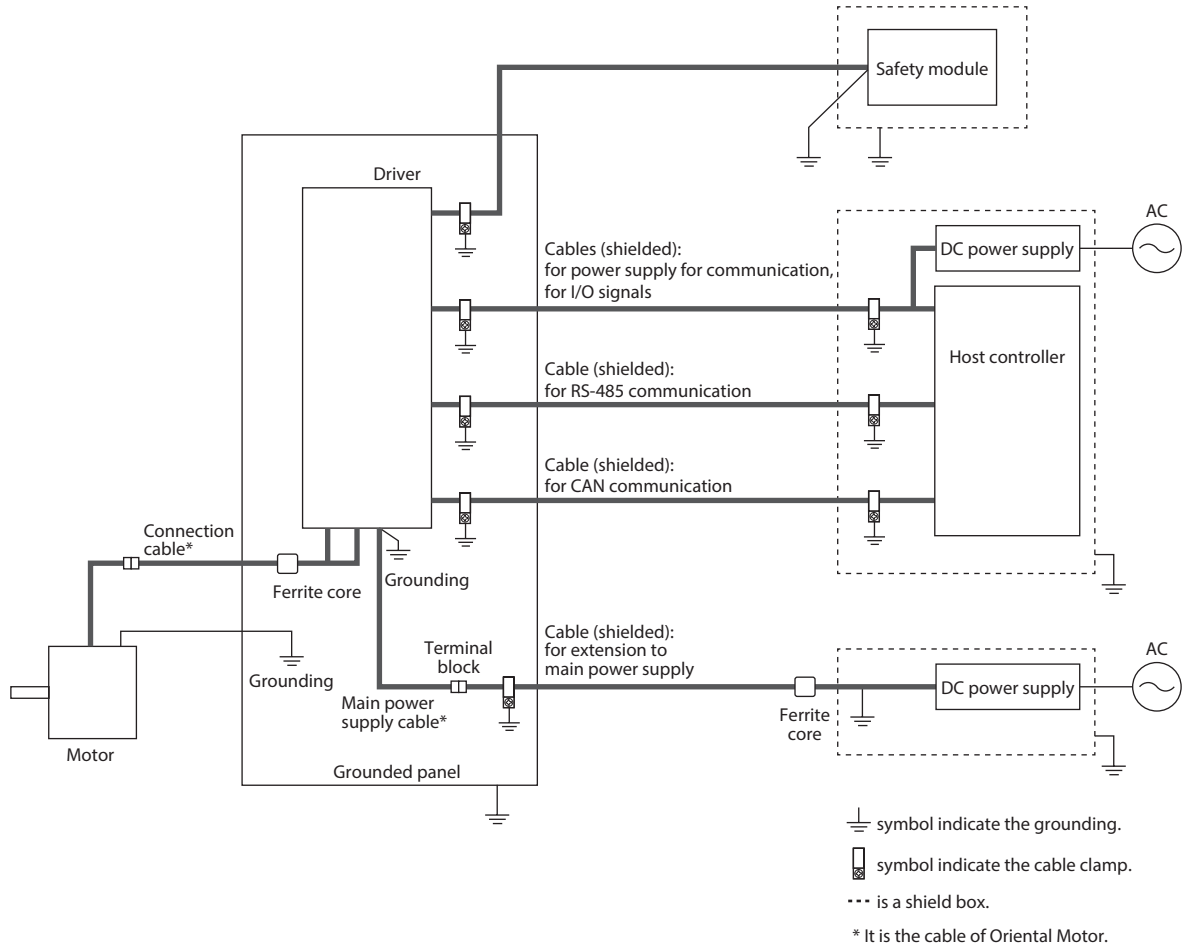
Oriental Motor products are compliant with the EMC Directive/Regulations in accordance with the contents of "6-7 Noise elimination measures" under the terms of "Example of installation and wiring". The results of EMC testing of the equipment will vary depending on the type, layout, wiring method, etc. of the parts and components used. Verify compliance with the EMC Directive/Regulations in a condition where all parts and components, including Oriental Motor products, are assembled into the equipment.



CAUTION

This equipment is not intended for use in residential environments nor for use on a low-voltage public network supplied in residential premises, and it may not provide adequate protection to radio reception interference in such environments.

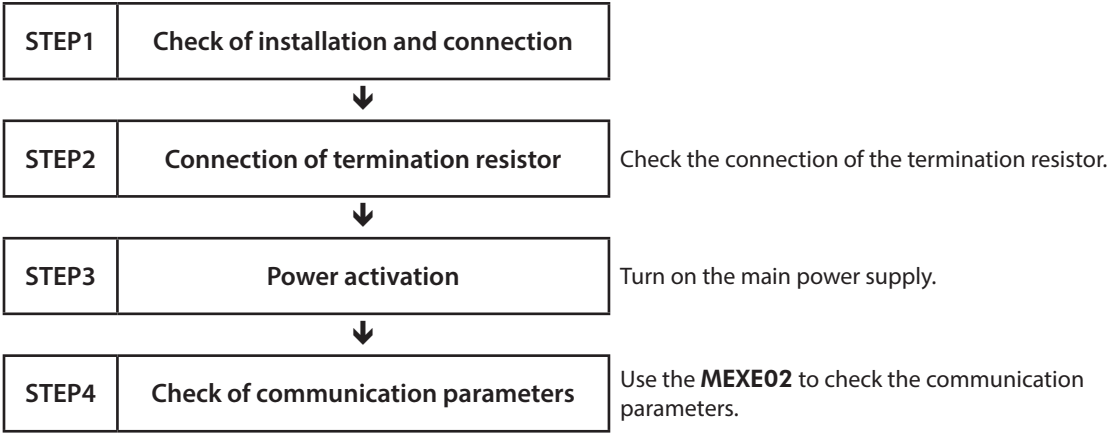
● Example of installation and wiring



The termination resistors are connected on both ends of the CAN communication cable to conduct the testing. The CAN-Bus cable is used for the CAN communication cable.

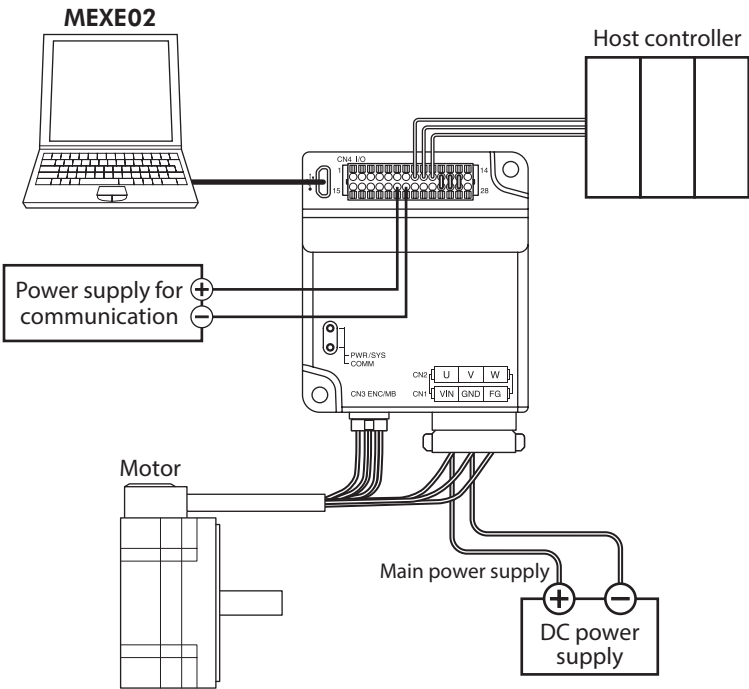
7 Guidance

If you are new to this product, read this section to understand the setting flow of the communication parameters.



7-1 Setting of RS-485 communication

STEP 1 Check of installation and connection



STEP 2 Connection of termination resistor

Connect a termination resistor for a driver located the farthest away (positioned at the end) from the host controller. There are the following two methods for how to connect a termination resistor.

● **When a termination resistor inside the driver is used**

Change the "RS-485 communication termination resistor" parameter to "Enable" with the **MEXE02**.

Name	Setting
RS-485 communication termination resistor	Enable

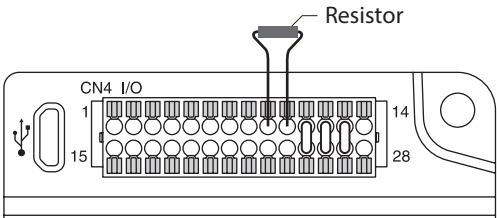
Note The termination resistor is turned ON only when the main power is supplied to the driver since it is turned ON or OFF inside the driver.

memo The termination resistor inside the driver is set based on the address number (server address).
The terminating resistor is enabled when the server address 4 is set (initial value).
When the server address 4 is used, check the connection of termination resistor.
Refer to the OPERATING MANUAL Function Edition for how to set the address number.

● **When a resistor (120 Ω) is connected between the TR+ and TR- terminals of the CN4 connector**

Connecting method

1. Connect lead wires to a resistor.
2. Connect the lead wires between the TR+ and TR- terminals of CN4.



- Note**
- Be sure to connect a resistor between the TR+ and TR- terminals. Incorrect connection may cause damage to the resistor.
 - When connecting a resistor, set the "RS-485 communication termination resistor" parameter to "Disable."

memo For a resistor, use a metal film resistor of 120 Ω, 1/2 W or more.

STEP 3 Power activation

Turn on the main power supply.
Start the **MEXE02**.
Execute "Communication port" to check the setting of the communication port.
Execute "Data reading" to read the driver data.

STEP 4 Check of communication parameters

Start "Starts the simple setting." of the **MEXE02**.

(m1) Communication setting

☒ Start the simple setting.

Communication function

COM Setting Condition CANopen / Modbus RTU

Power supply for communication * To communicate, the communication power supply is required to be turned on.
Communication power supply lost action:

ID-SEL(Applicable value) ☐ ID-SEL0 ☐ ID-SEL1 ☐ ID-SEL2 ☐ ID-SEL3
ID-SEL(Present value) ☐ ID-SEL0 ☐ ID-SEL1 ☐ ID-SEL2 ☐ ID-SEL3

CANopen Communication setting

Node-ID Reflecting on the driver.
Baudrate

☒ Statusword - remote CANopen com. status

Communication status
Communication error
Reception count
Transmission count

Modbus Communication setting

Slave address Reflecting on the driver.
Baudrate
Communication parity
Termination resistor RS-485 com. status

Communication stop bit Communication timeout [ms]
Transmission waiting time [ms] Silent interval [ms]
Communication error detection

Communication error condition
Reception byte Transmission byte
Normal reception frame (Only own) Transmission frame
Communication interval [ms]

Set the following communication parameters according to the communication parameters of the host controller.

Modbus Communication setting

Slave address Reflecting on the driver.
Baudrate
Communication parity
Termination resistor RS-485 com. status

If the values are different, change the value of the "Input value" and execute "Reflecting on the driver."

If the following communication parameters are different from those of the host controller, execute "Detailed setting..." to change the parameters.

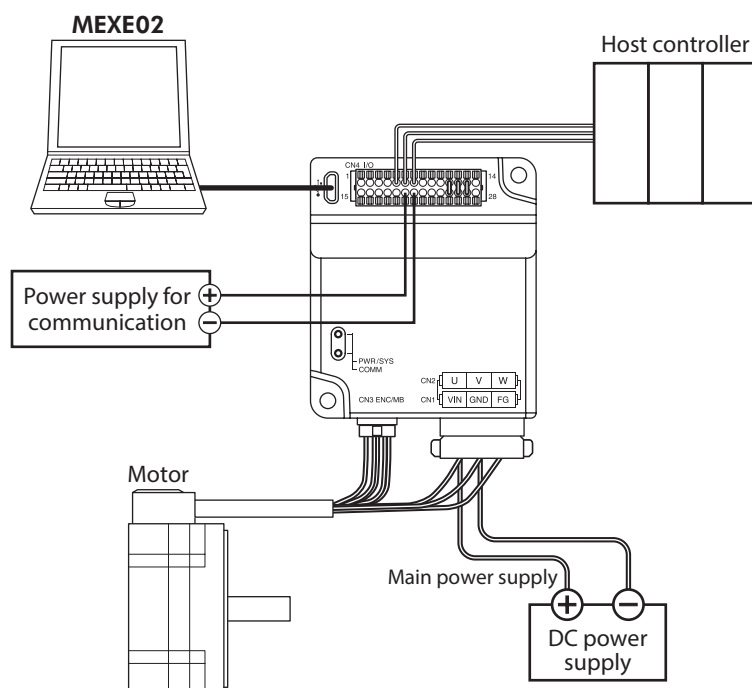
Parameter name	Initial value
Byte & word order (Modbus)	Even Address-High Word & Big-Endian
Communication stop bit (Modbus)	1 bit



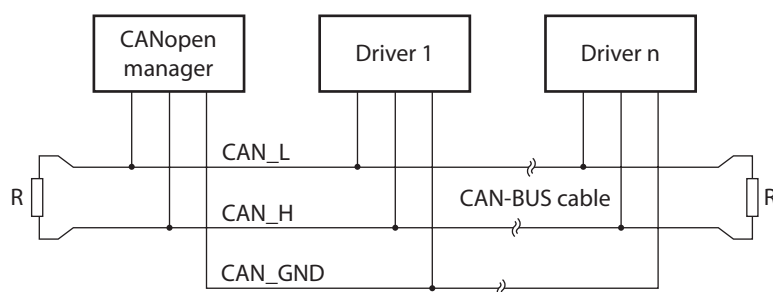
The "Slave address" can be changed by the ID-SEL0 and ID-SEL1 inputs.

7-2 Setting of CAN communication

STEP 1 Check of installation and connection



STEP 2 Connection of termination resistor



R: Termination resistor

Connect the termination resistor (120 Ω , 1/4 W or more) on both ends of a bus. Termination resistors are not included with the product.

STEP 3 Power activation

Turn on the main power supply.

Start the **MEXE02**.

Execute "Communication port" to check the setting of the communication port.

Execute "Data reading" to read the driver data.

STEP 4 Check of communication parameters

Start "Starts the simple setting." of the **MEXE02**.

(m1) Communication setting

☒ Start the simple setting.

Communication function

COM Setting Condition CANopen / Modbus RTU

Power supply for communication * To communicate, the communication power supply is required to be turned on.
Communication power supply lost action:

ID-SEL(Applicable value) ☐ ID-SEL0 ☐ ID-SEL1 ☐ ID-SEL2 ☐ ID-SEL3
ID-SEL(Present value) ☐ ID-SEL0 ☐ ID-SEL1 ☐ ID-SEL2 ☐ ID-SEL3

CANopen Communication setting

	Input value	Present value	
Node-ID	Follow ID-SEL input	1	Reflecting on the driver.
Bitrate	500 kbps	500 kbps	
☒ Statusword - remote			
Communication status	Initialization		CANopen com. status
Communication error	No error		
Reception count	0		
Transmission count	0		

Modbus Communication setting

	Input value	Present value	
Slave address	Follow ID-SEL input	1	Reflecting on the driver.
Baudrate	230400 bps	230400 bps	
Communication parity	Even	Even	RS-485 com. status
Termination resistor	Enable When Slave address=4	Disable	
Communication stop bit	1 bit	Communication timeout [ms] 0	
Transmission waiting time [ms]	3.0	Silent interval [ms] 0.0	
Communication error detection	3		
Communication error condition	00:Communication error not present		
Reception byte	0	Transmission byte 0	
Normal reception frame (Only own)	0	Transmission frame 0	
Communication interval [ms]	0		

Set the following communication parameters according to the communication parameters of the host controller.

CANopen Communication setting

	Input value	Present value	
Node-ID	Follow ID-SEL input	1	Reflecting on the driver.
Bitrate	500 kbps	500 kbps	
☒ Statusword - remote			
Communication status	Initialization		CANopen com. status
Communication error	No error		
Reception count	0		
Transmission count	0		

If the values are different, change the value of the "Input value" and execute "Reflecting on the driver."



The "Node-ID" can be changed by the ID-SEL0 and ID-SEL1 inputs.

8 Inspection and maintenance

8-1 Inspection

It is recommended that periodic inspections are conducted for the items listed below after each operation. If an abnormality is generated, discontinue any use and contact your nearest Oriental Motor sales office.

■ Inspection item

- Check if any of the mounting screws secured the driver is loose.
- Check if the connection part with the connector is loose.
- Check if dust is deposited on the driver.
- Check if the driver has unusual smells or appearance defects.



The driver uses semiconductor components. Static electricity may damage the semiconductor components of the driver, so be extremely careful when handling them.

8-2 Warranty

■ Product warranty

Check on the Oriental Motor Website for the product warranty.

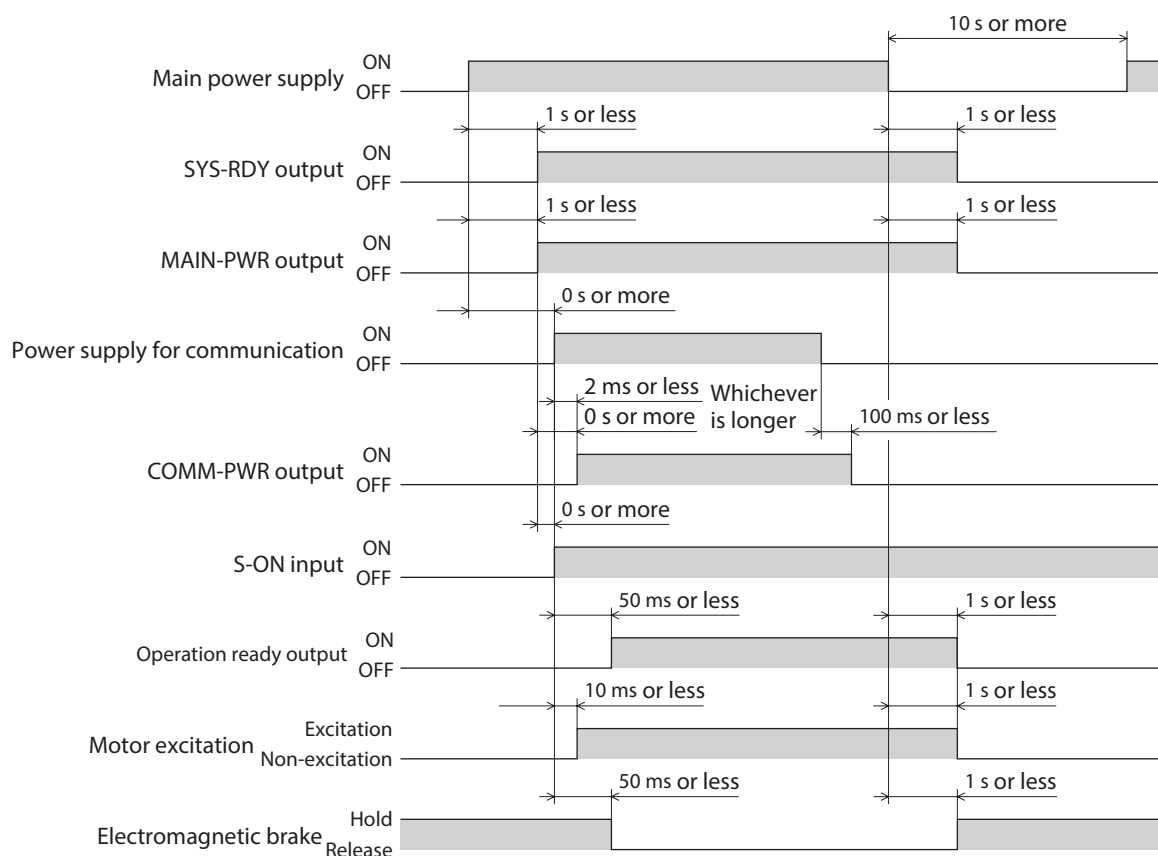
8-3 Disposal

Dispose the product correctly in accordance with laws and regulations, or instructions of local governments.

9 Appendix

9-1 Timing chart

■ Power activation



9-2 Alarm list

Alarm code	Number of LED blinks	Alarm type	Reset using the ALM-RST input	Motor excitation*1
10h	7	Position deviation	Possible	Non-excitation after deceleration
20h	9	Overcurrent	Not possible	Non-excitation
21h	7	Main circuit overheat	Possible	Non-excitation after deceleration
22h	5	Overvoltage		Non-excitation
25h		Undervoltage		Non-excitation after deceleration
26h	7	Motor overheat		
28h	2	Encoder error	Not possible	Non-excitation
29h	9	Internal circuit error		
2Ah	2	Encoder communication error		
2Dh	2	Motor connection error*2		

Alarm code	Number of LED blinks	Alarm type	Reset using the ALM-RST input	Motor excitation*1
30h	7	Overload	Possible	Non-excitation after deceleration
31h		Overspeed		
41h	9	EEPROM error	Not possible	Non-excitation
42h	2	Initial encoder error		
44h	9	Encoder EEPROM error		
45h	2	Motor combination error		
4Ah	7	Return-to-home incomplete	Possible	Excitation
50h	9	Electromagnetic brake overcurrent	Not possible	Non-excitation
53h	3	HWTIO input circuit error		
55h	2	Electromagnetic brake connection error		
60h	3	±LS both sides active	Possible	Excitation
61h		Reverse ±LS connection		
62h	4	Return-to-home operation error		
63h		No HOMES		
64h		Z, SLIT signal error		
66h		Hardware overtravel		
67h	6	Software overtravel		
68h	1	HWTIO input detection		Non-excitation
6Ah	6	Return-to-home additional operation error		Excitation
6Eh	1	User alarm*2		Non-excitation after deceleration*3
70h	6	Operation data error	Not possible	Excitation
71h		Unit setting error		Non-excitation
81h	8	Network bus error	Possible	Excitation
84h		RS-485 communication error		
85h		RS-485 communication timeout		
8Ch		Outside setting range	Not possible	Non-excitation
F0h	Lighting	CPU error		
F3h	6	CPU overload		

*1 An excitation state of the motor when an alarm is generated is as follows.

Non-excitation: If an alarm is generated, the motor current is cut off and the motor holding force is lost.

When an electromagnetic brake motor is used, the electromagnetic brake automatically actuates to hold the motor shaft.

Non-excitation after deceleration: If an alarm is generated, the motor decelerates to a stop.

After decelerating to a stop, the motor current is cut off and the motor holding force is lost.

When an electromagnetic brake motor is used, the electromagnetic brake automatically actuates to hold the motor shaft.

Excitation: If an alarm is generated, the motor will decelerates to a stop.

After decelerating to a stop, the motor current is not shut off and the motor excitation state is continued.

*2 It is effective for the driver version 3.00 or later.

*3 This is the initial setting. The motor excitation state after stop can be set with the "User alarm action" parameter.

9-3 Specifications

■ Specifications

Motor models in the table below describe a part of the entire name of models. Refer to p.9 for models in details.

Model	Driver	BLVD-KRD			
	Motor	BLMR260HK BLMR460SHK	BLMR5100K	BLMR5200K BLMR6200SK	BLMR5400K BLMR6400SK
Rated output power		60 W	100 W	200 W	400 W
Power supply input	Rated voltage	24 to 48 VDC			48 VDC
	Allowable operating voltage	15 to 55 VDC			30 to 55 VDC
	Rated current	1.7 A (48 VDC) to 3.3 A (24 VDC)	2.6 A (48 VDC) to 5.1 A (24 VDC)	5.3 A (48 VDC) to 10.5 A (24 VDC)	10.4 A
	Maximum input current	5.5 A	10 A	18 A	16 A
Power supply input for communication	Rated voltage	24 to 48 VDC			
	Allowable operating voltage	15 to 55 VDC			
	Maximum input current	0.2 A			
I/O signals	Input signal	12 to 30 VDC			
	Output signals	12 to 30 VDC Current 10 mA or less			
Rated torque		0.191 N·m (1.69 lb-in)	0.319 N·m (2.8 lb-in)	0.637 N·m (5.6 lb-in)	1.27 N·m (11.2 lb-in)
Peak torque		0.382 N·m (3.3 lb-in) [200 %]	0.704 N·m (6.2 lb-in) [220 %]	1.34 N·m (11.8 lb-in) [210 %]	2.54 N·m (22 lb-in) [200 %]
Rated speed		3000 r/min			
Speed control range		1 to 4000 r/min (Speed ratio 1:4000)			
Time rating		Continuous			30 minutes*

* Operation can be performed for 30 minutes continuously. However, operation for more than 30 minutes can be performed depending on an ambient temperature and heat radiation condition.

■ General specifications

Operating environment	Ambient temperature	0 to +40 °C (+32 to +104 °F) (non-freezing)
	Humidity	85 % or less (non-condensing)
	Altitude	Up to 1000 m (3300 ft.) above sea level
	Surrounding atmosphere	No corrosive gas, dust, water or oil. Cannot be used in radioactive materials, magnetic field, vacuum or other special environments.
	Vibration	Not subject to continuous vibration or excessive impact. In conformance with JIS C 60068-2-6 "Sine-wave vibration test method" Frequency range: 10 Hz to 55 Hz Pulsating amplitude: 0.15 mm (0.006 in.) Sweep direction: 3 directions (X, Y, Z) Number of sweeps: 20 times
Storage environment Shipping environment	Ambient temperature	−25 to +70 °C [−13 to +158 °F] (non-freezing)
	Humidity	85 % or less (non-condensing)
	Altitude	Up to 3000 m (10000 ft.) above sea level
	Surrounding atmosphere	No corrosive gas, dust, water or oil. Cannot be used in radioactive materials, magnetic field, vacuum or other special environments.
Degree of protection		IP20

■ RS-485 communication specifications

Electrical characteristics	In conformance with EIA-485 Use twisted-pair wires and keep the total extension distance up to 10 m (32.8 ft.). *
Communication mode	Half duplex Asynchronous mode (data: 8 bits, stop bit: 1 bit/2 bits, parity: none/even number/odd number)
Transmission rate	Selectable from 9,600 bps, 19,200 bps, 38,400 bps, 57,600 bps, 115,200 bps, and 230,400 bps (initial value).
Protocol	Modbus RTU mode
Type of Connection	Up to 31 drivers can be connected to one host controller.

* If the motor cable or power supply cable generates an undesirable amount of noise depending on the wiring or configuration, shield the cable or install a ferrite core.

■ CAN communication specifications

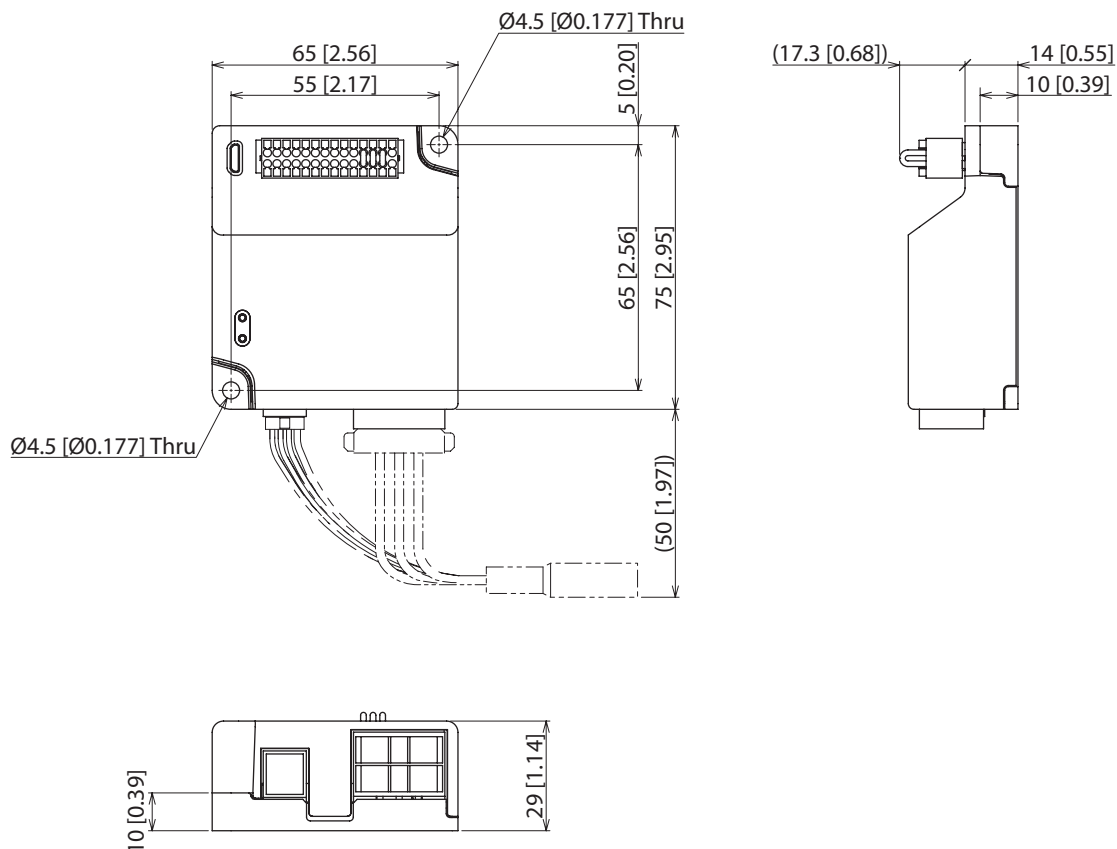
Electrical characteristics	In conformance with ISO 11898 Use the CAN-Bus cable.
Communication protocol	CANopen
Communication profile	In conformance with CiA DS301 Version 4.2.0
Device profile	In conformance with CiA DSP402 Version 4.0.0
Node ID	1 to 127
Bit rate	Selectable from 1 Mbps, 800 kbps, 500 kbps (initial value), 250 kbps, 125 kbps, 50 kbps, 20 kbps, 10 kbps
Maximum bus length	25 m (82 ft.) [maximum bus length at 1 Mbps]
Communication objects	NMT (Network Management) SDO (Service Data Object: 1 SDO server) PDO (Process Data Object: 4 Receive-PDO, 4 Transmit-PDO) EMCY (Emergency Object) SYNC (Synchronization Object)
Operation modes	Profile velocity mode (pv) Profile position mode (pp) Profile torque mode (tq)* Homing mode (hm)

* It is effective for the driver version 4.00 or later.

■ Dimensions

Mass: 0.12 kg (0.26 lb.)

[Unit: mm (in.)]



9-4 Regulations and standards

■ **UL Standards, CSA Standards**

This product is recognized by UL under the UL and CSA Standards.

■ **CE Marking/UKCA Marking**

This product is affixed with the marks under the following directives/regulations.
Declaration of Conformity can be downloaded from Download Page of the product in Oriental Motor Website (<https://www.orientalmotor.eu/>).

● **EU EMC Directive/UK EMC Regulation**

Refer to "6-8 Compliance with the EMC Directive/Regulations" on p.26 for details about conformity.

● **EU Machinery Directive /UK Machinery Regulation**

Applicable standards: EN ISO 12100, EN 61800-5-2, EN ISO 13849-1:2023

● **EU RoHS Directive/UK RoHS Regulation**

This products do not contain the substances exceeding the restriction values.

■ **Functional safety**

This product is certified by TÜV SÜD Product Service GmbH under the following standards and affixed with the TÜV SÜD Mark. It is not a certified product if the TÜV SÜD Mark is not affixed.

Applicable standards	Functional safety	IEC 61800-5-2, EN 61800-5-2
		IEC 61508-1, EN 61508-1
	Electrical safety	IEC 61508-2, EN 61508-2
		ISO 13849-1:2023, EN ISO 13849-1:2023
	EMC	EN 61800-5-1
		IEC 61000-6-7, EN 61000-6-7
Safety function		STO (Safe Torque Off)

The driver is not provided with the motor overtemperature protection specified in EN Standards.
The driver is provided with the electronic motor overload protection specified in EN Standards.
Refer to "Function Edition" for detection of the overload alarm.



The driver is provided with the electronic motor overload protection, but is not provided with the thermal memory retention function and the speed sensitive function.

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